

Project Title:

Online Programming Contest Management System

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1. Introduction and System Description

The proposed system is a web-based platform designed to manage and organize programming contests. It allows users to register, participate in contests, solve problems, and track their performance through rankings and submission history.

The system aims to provide a centralized environment for competitive programming activities. It supports both internally created problems and externally sourced problems, enabling a diverse and flexible problem set for contests. Additionally, users can contribute by creating and submitting their own problems to the system.

2. System Functionality

The system will perform the following functions:

- User registration and authentication
- Creation and management of programming contests
- Addition and management of problems
- Inclusion of problems from external platforms (stored as references)
- Allowing users to create and submit their own problems
- Assigning multiple problems to contests
- Allowing users to participate in contests
- Submission of solutions by users
- Storage and tracking of all submissions
- Generation of leaderboards based on user performance
- Viewing submission history and contest details

3. Users of the System

Administrator

- Create, update, and delete contests
- Manage all problems (internal and external)
- Approve or moderate user-created problems
- Assign problems to contests
- Monitor user activity and submissions
- Manage system data

Registered User (Participant)

- Register and log into the system
- View available contests
- Participate in contests
- View problems assigned to contests
- Submit solutions
- View submission results and rankings
- Create and submit new problems

4. Technologies Used

Front-End:

- HTML
- CSS
- JavaScript
- Bootstrap

Back-End:

- Node.js / Python (Flask or Django)

Database:

- MySQL / PostgreSQL

5. Scope of the System

Included:

- Contest creation and management
- Problem management (internal and external)
- User-generated problem submission and moderation
- User participation tracking
- Submission handling and storage
- Leaderboard generation
- User performance tracking

Not Included:

- Real-time code execution engine
- Direct API integration with external platforms
- Advanced plagiarism detection

6. Assumptions (Optional)

- Each user will have a unique account
- Problems can either be internally created or externally referenced
- External problems will be stored with metadata (e.g., source, link)
- User-created problems may require approval before use in contests
- Each submission will be evaluated and assigned a status
- A user can submit multiple solutions for a problem
- Each contest has a fixed start and end time